# A closed form for the $p$-adic valuation of OEIS A129365, and a proof that its terms are integers

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**Abstract**

OEIS A129365 is the quotient $a(n) = \left(\prod_{j,k\le n} \gcd(j,k)\right)/\left(\prod_{k=1}^{n}(\lfloor n/k\rfloor!)^k\right)$. In April 2007 P. Bala listed four conjectures about it, the first being simply that $a(n)$ is always an integer. We prove the exact closed form

$$\operatorname{ord}_p a(n) = \sum_{i\ge 1} B\big(\lfloor n/p^i\rfloor\big), \qquad B(M) = \sum_{k=1}^{M}\big(M \bmod k\big) \quad \text{(A004125)},$$

for every prime $p$. Since a sum of remainders is nonnegative, every exponent is nonnegative and Conjecture A follows at once. The proof is three steps: the known factorisation of the numerator, the nested-floor identity $\lfloor n/(kp^i)\rfloor = \lfloor\lfloor n/p^i\rfloor/k\rfloor$ for the denominator, and the observation that $M^2 - \sum_{k\le M} k\lfloor M/k\rfloor$ is exactly $\sum_{k\le M}(M \bmod k)$ because $M^2 = \sum_{k\le M} M$.

## 1 The sequence and the conjecture

OEIS A129365 (Peter Bala, 13 April 2007) has offset 1 and is defined by $a(n) = \text{A092287}(n)/\text{A129364}(n)$, equivalently by the formula recorded on the entry

$$a(n) = \frac{\displaystyle\prod_{j=1}^{n}\prod_{k=1}^{n}\gcd(j,k)}{\displaystyle\prod_{k=1}^{n}\big(\lfloor n/k\rfloor!\big)^k}, \tag{1}$$

with $a(1),\dots,a(10) = 1,1,1,1,1,2,2,2,6,48$. The entry carries four conjectures, labelled A–D. This note proves the first:

**Conjectures:** A) $a(n)$ is always an integer.

Conjectures B and C were addressed by T. Adamczewski (arXiv:2608.11941, August 2026), as annotated on the entry by M. De Vlieger on 20 August 2026. Conjecture D is the same closed form evaluated at $n \mapsto np$. As of the "Last modified" line on the live entry (23 August 2026), A and D are still recorded as unproven.

Throughout $p$ is a prime, $v_p(N)$ is the exponent of $p$ in $N$ (the entry's $\operatorname{ord}_p$), and

$$B(M) = \sum_{k=1}^{M}\big(M \bmod k\big)$$

is A004125, the sum of remainders, with $B(0) = 0$.

## 2 Three ingredients

**Lemma 1** (numerator). $v_p\Big(\prod_{j=1}^{n}\prod_{k=1}^{n}\gcd(j,k)\Big) = \sum_{i\ge 1}\lfloor n/p^i\rfloor^2.$

*Proof.* $v_p(\gcd(j,k)) = \min(v_p(j), v_p(k))$ counts the $i \ge 1$ with $p^i \mid j$ and $p^i \mid k$. Summing over $j,k \le n$ and exchanging the order of summation, the two divisibility conditions are independent, so the count at level $i$ is $\big(\#\{j \le n : p^i \mid j\}\big)^2 = \lfloor n/p^i\rfloor^2$. (This is the square-case formula of A092287, confirmed on that entry by C. R. Greathouse IV in April 2013.) □

**Lemma 2** (nested floors). *For positive integers $n, k, q$, $\lfloor n/(kq)\rfloor = \lfloor\lfloor n/q\rfloor/k\rfloor$.*

*Proof.* Standard: for a real $x$ and a positive integer $k$, $\lfloor x/k\rfloor = \lfloor\lfloor x\rfloor/k\rfloor$; apply it to $x = n/q$. □

**Lemma 3** (remainder identity). *For every integer $M \ge 0$, $M^2 - \sum_{k=1}^{M} k\lfloor M/k\rfloor = \sum_{k=1}^{M}\big(M \bmod k\big) = B(M)$.*

*Proof.* $M^2 = \sum_{k=1}^{M} M$, and $M - k\lfloor M/k\rfloor = M \bmod k$ termwise. □

## 3 The closed form

**Theorem 4.** *For every $n \ge 1$ and every prime $p$,*

$$v_p\big(a(n)\big) = \sum_{i\ge 1} B\big(\lfloor n/p^i\rfloor\big),$$

*where $a(n)$ is given by (1). All terms with $p^i > n$ vanish, so the sum is finite.*

*Proof.* Write $M_i = \lfloor n/p^i\rfloor$. By Legendre's formula and Lemma 2,

$$v_p\Big(\prod_{k=1}^{n}\big(\lfloor n/k\rfloor!\big)^k\Big) = \sum_{k=1}^{n}\sum_{i\ge 1} k\big\lfloor\lfloor n/k\rfloor/p^i\big\rfloor = \sum_{i\ge 1}\sum_{k=1}^{n} k\big\lfloor n/(kp^i)\big\rfloor = \sum_{i\ge 1}\sum_{k=1}^{M_i} k\lfloor M_i/k\rfloor,$$

where in the last step Lemma 2 was used again in the form $\lfloor n/(kp^i)\rfloor = \lfloor M_i/k\rfloor$, and the terms with $k > M_i$ were dropped because $\lfloor M_i/k\rfloor = 0$ there. Subtracting this from Lemma 1 and applying Lemma 3 at each level $i$,

$$v_p\big(a(n)\big) = \sum_{i\ge 1}\Big(M_i^2 - \sum_{k=1}^{M_i} k\lfloor M_i/k\rfloor\Big) = \sum_{i\ge 1} B(M_i). \qquad \square$$

**Corollary 5** (Conjecture A). *$a(n)$ is an integer for every $n \ge 1$.*

*Proof.* Each $M \bmod k$ is a nonnegative integer, so $B(M) \ge 0$ and hence $v_p(a(n)) \ge 0$ for every prime $p$ by Theorem 4. A positive rational with nonnegative valuation at every prime is an integer. □

**Remark** (the other three conjectures). Theorem 4 settles all four of Bala's conjectures at once, though only A and D were still open. For B: $B(M) = 0$ exactly when $M \le 2$ (for $M \ge 3$ the term $k = 2$ or $k = M - 1$ contributes), so $v_p(a(n)) > 0$ iff $\lfloor n/p\rfloor \ge 3$ iff $p \le n/3$. For C: if $m = np + r$ with $0 \le r < p$ then $\lfloor m/p^i\rfloor = \lfloor n/p^{i-1}\rfloor$ for every $i \ge 1$, independent of $r$, so $v_p(a(m))$ is constant across the block. For D: taking $m = np$ in that same identity gives $v_p(a(np)) = B(n) + B(\lfloor n/p\rfloor) + B(\lfloor n/p^2\rfloor) + \cdots$. B and C are credited on the entry to Adamczewski (2026); the derivations above are recorded only to show that the closed form covers them.

**Remark** (honest assessment)**.** The proof is short and uses nothing beyond Legendre's formula and two elementary floor identities. It is an OEIS comment, not a paper. What makes it worth writing down is that the closed form is not on the entry in any form, and that it replaces four separate conjectures with one identity.

One caution the reader should carry: a benchmark of 492 Lean-formalised open OEIS conjectures is currently being worked through by automated systems (Adamczewski, arXiv:2608.11941, 12 August 2026, reporting 147 resolved; and earlier work on the same 492). Conjectures B and C of this entry fell to that effort three days before this note was written. I have not been able to confirm whether A or D appear in that formalised set, and anyone acting on this note should check the results repositories before assuming priority.

## 4 Verification

A verification script, written separately from this note, checks every claim and prints every number quoted. It (i) recomputes $a(1), \ldots, a(27)$ from (1) and matches the published DATA of A129365 exactly; (ii) confirms $B(M) = 0 \iff M \leq 2$ for $0 \leq M \leq 2000$, and prints $B(0), \ldots, B(12) = 0, 0, 0, 1, 1, 4, 3, 8, 8, 12, 13, 22, 17$; (iii) compares $v_p(a(n))$ against $\sum_i B(\lfloor n/p^i \rfloor)$ for every $1 \leq n \leq 60$ and every prime $p \leq n$, that is 597 pairs $(n, p)$, with 0 mismatches; and (iv) confirms directly that $a(n)$ is an integer for $1 \leq n \leq 100$.

## References


[1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, `https://oeis.org/A129365`, entry A129365, consulted 23 August 2026; also entries A092287 and A004125.

[2] A. M. Legendre, *Théorie des Nombres*, 3rd ed., Firmin Didot, Paris, 1830.

[3] T. Adamczewski, *OEIS Open: How many conjectures can language models turn into theorems?*, arXiv:2608.11941 [cs.AI], 2026.